

STAT656 Final Project:

Clustering and Profiling Spotify Songs

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I. Abstract

This project discusses the structure of music by analyzing its characteristics in a large-scale collection of digital music. Using a dataset of over 130,000 songs from Spotify, we employ multiple unsupervised learning techniques, specifically Principal Component Analysis (PCA), and K-Means clustering, Hierarchical Clustering, and Autoencoder through Pytorch to identify distinct “audio profiles”. After filtering for established popularity, we analyze nine features, like danceability, energy, and acousticness, to create groups of songs based on their similarities. The goal of this project is to determine how technical attributes correlate with popularity allowing for the analysis of modern music distribution centered around similarities rather than traditional genres.

I. Introduction

In the era of streaming, the division of music has shifted from genres to data based exploration. Spotify, one of the world’s leading streaming programs, quantifies the technical aspects of music through audio features that describe a track’s mood, energy, and rhythm. Understanding the interactions between these features is the key to recommendation programs and musicology.

This study performs exploratory data analysis (EDA) and cluster analysis on Spotify’s 2019 data. The primary objective of this is to move beyond manual categorization using K-Means clustering to explore natural groups within music. Using PCA to reduce dimensionality of the data, we can visualize these clusters and assess the relationship between the technical and commercial attributes.

II. Data Preprocessing and Cleaning

The analysis uses a dataset from Spotify that consists of over 130,000 tracks and 17 variables, including categorical and quantitative variables. For simplicity, this data was pared down to consist of tracks with a popularity greater than 0, leaving about 111,800 tracks. In addition, the nine audio features in Table 1, below were selected for clustering. Finally, before analysis and after EDA we used a standard scaler to get means of 0 and sd of 1 for each variable and then transformed the data to fit that.

	danceability	energy	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo
count	111774	111774	111774	111774	111774	111774	111774	111774	111774
mean	0.577959	0.569467	-9.831834	0.111265	0.344021	0.206224	0.194912	0.435165	119.47424
std	0.188435	0.260992	6.545899	0.123257	0.347	0.348923	0.167262	0.255739	30.155886
min	0	0	-60	0	0	0	0	0	0
25%	0.454	0.395	-11.609	0.0388	0.0326	0	0.0975	0.222	95.89975
50%	0.599	0.606	-7.785	0.0555	0.203	0.000094	0.124	0.415	120.014
75%	0.722	0.777	-5.58	0.128	0.642	0.287	0.237	0.63	139.869
max	0.996	1	1.806	0.966	0.996	1	0.999	1	249.983

Table 1: Summary Statistics of Audio Features

III. Exploratory Data Analysis

A. Statistical Analysis

These highlight substantial variation in both scale and distribution across these variables. For example, loudness has a wide range (from -60 to 1.806), while features such as danceability and energy are bounded between 0 and 1. Additionally, the relatively large standard deviations for variables like acousticness (0.347) and instrumentalness (0.349) indicate high variability across songs. Additionally, it is clear that many features are not symmetrically distributed. The large differences between the 25th, 50th, and 75th percentiles—particularly for acousticness and instrumentalness—suggest skewness in the data. These observed patterns reinforce our use of feature scaling prior to clustering, as variables with larger ranges or variance would otherwise dominate the distance calculations used in clustering algorithms.

B. Distribution Analysis

The distributions of the audio features, shown in Figure 1, further confirm the variability observed in Table 1. Acousticness and instrumentalness exhibit strong right-skewness, with a large concentration of songs near zero and a smaller number of songs with very high values. This indicates that most songs are not highly acoustic or instrumental, but a subset forms a distinct group. In contrast, energy and loudness appear more concentrated toward higher values, suggesting that a large portion of songs in the dataset are relatively intense.

The variation in these distributions supports the idea that songs differ across multiple independent dimensions. Additionally the patterns suggest that clusters are likely to form based on combinations of these features—for example, separating high-energy songs from more acoustic or instrumental tracks which again motivated us to use clustering techniques to identify such groupings.

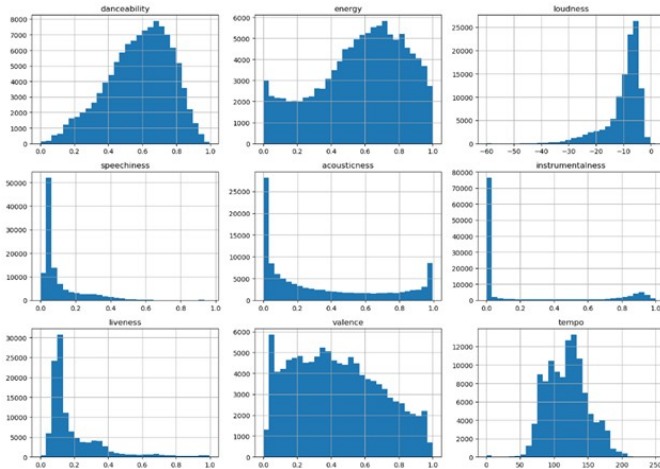


Figure 1: Distribution of audio features showing skewness and variability across variables

C. Correlation Analysis

To further explore the data we used a correlation heatmap in Figure 2 which provides additional insight into relationships between features. A strong positive correlation between energy and loudness is clearly visible, confirming that louder songs tend to be more energetic. A strong negative relationship between acousticness and energy can also be seen meaning more acoustic heavy songs are not as energetic. Moderate relationships, such as between danceability and valence, are also present, while many other features show relatively weak correlations.

These patterns indicate that while some variables capture overlapping information, many contribute unique variation to the dataset. This is important for clustering, as it suggests that meaningful groupings will emerge from combinations of features rather than redundancy.

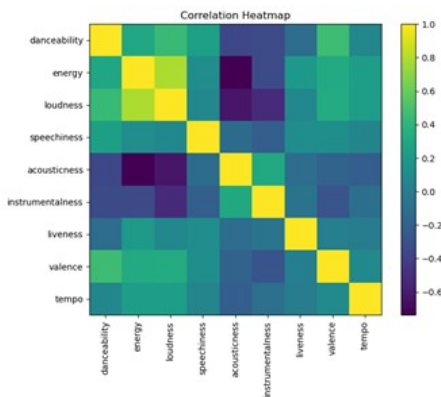


Figure 2: Correlation heatmap illustrating relationships between audio features.

D. Patterns and Key Insights

The EDA reveals several important patterns in the dataset. First, no single feature fully explains song structure, meaning that clustering must rely on combinations of variables. Second, songs vary across multiple independent dimensions, including intensity, acoustic properties, and rhythm. Based on these observations, likely cluster structures include high-energy versus low-energy songs, acoustic versus electronic compositions, and instrumental versus vocal tracks.

Overall, the dataset demonstrates sufficient variability and structure to justify clustering as well as dimensionality reduction techniques, such as PCA Principal Component Analysis (PCA), to better capture underlying patterns and improve clustering performance.

IV. Principal Component Analysis (PCA)

As stated from the EDA, it was clear that Principal Component Analysis (PCA) was needed and applied to reduce the dimensionality of the dataset while preserving its key structure. As shown in Figure 3, the PCA projection captures the main variation in the data, allowing songs to be visualized in a lower-dimensional space. The clustering tendencies visible in this projection suggest that meaningful groupings already exist in the transformed feature space.

Additionally, the PCA results indicate that most of the dataset's variability can be explained by a relatively small number of components (approximately 8), enabling dimensionality reduction with minimal information loss. This was able to improve both computational efficiency and the effectiveness of clustering methods applied in later sections.

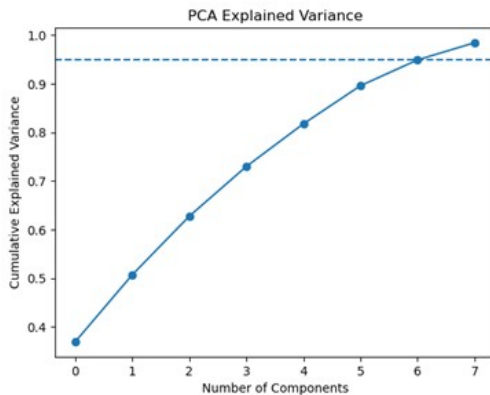


Figure 3: PCA projection of the dataset showing the main variation in song features.

V. Analytic Reasoning

Given what was discovered in the EDA. The statistical analysis in this project is centered on the application of clustering methods. K-means clustering was used as the primary baseline model due to its strong statistical foundation in minimizing within-cluster sum of squares and its efficiency for large datasets. This approach was applied to PCA-transformed data to ensure that

clustering was performed in a reduced, orthogonal feature space, addressing issues of multicollinearity and noise. Hierarchical clustering using Ward’s linkage was also implemented to provide an alternative perspective on the structure of the data, leveraging its variance-minimization properties to build clusters iteratively. Although hierarchical clustering was limited to a sample due to computational constraints, it served as an important exploratory tool to validate cluster structure and support the selection of the number of clusters.

To extend beyond linear methods, a deep learning approach was incorporated through the use of an autoencoder implemented in PyTorch. The autoencoder learns a nonlinear transformation of the data by compressing input features into a lower-dimensional latent space and reconstructing them, effectively capturing complex relationships that traditional methods may not detect. Clustering was then performed on the learned latent features using K-means, resulting in improved separation and cohesion of clusters. Model performance across all approaches was evaluated using internal validation metrics, including the Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index. These metrics provided a consistent and statistically grounded basis for comparison.

A. K-Means Clustering

K-means clustering results, shown in Figure 4 and summarized in Table 2, demonstrate clear separation of songs into distinct groups. The table highlights meaningful differences between clusters. For instance, Cluster 0 contains songs with high danceability and valence, while Cluster 3 represents songs with very low energy and high acousticness and instrumentality. Additionally, average popularity varies across clusters, with Cluster 0 showing the highest values and Cluster 3 the lowest, suggesting that certain audio characteristics may be associated with more popular songs.

These differences confirm that K-means successfully partitions the dataset into interpretable groups based on audio features. The visual separation in Figure 4 further supports this conclusion, as clusters occupy distinct regions in PCA space. However, some overlap remains, indicating that more complex relationships may not be fully captured by this method.

cluster	danceability	energy	loudness	speechiness	acousticness	instrumentality	liveness	valence	tempo	avg_popularity
0	0.703874	0.697552	-6.447102	0.080316	0.178064	0.078683	0.147459	0.627364	120.092816	31.743323
1	0.687764	0.609391	-8.723502	0.367671	0.252945	0.034003	0.224571	0.499543	121.608312	28.599777
2	0.553048	0.375028	-11.147223	0.059423	0.633185	0.148796	0.151212	0.351107	113.468874	28.901621
3	0.3277	0.152129	-22.559329	0.051807	0.876724	0.741799	0.153204	0.212173	102.508236	18.371623
4	0.490514	0.769782	-6.833542	0.07717	0.082102	0.23369	0.311075	0.310167	132.624704	28.376207

Table 2: K-Means Clustering Audio Features Breakdown

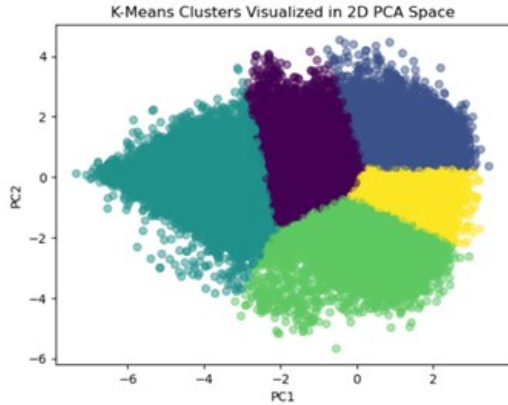


Figure 4: K-means clustering results visualized in PCA space

B. Hierarchical Clustering

Hierarchical clustering results provide additional insight into the structure of the data. The dendrogram in Figure 5 shows how observations are progressively merged into clusters, illustrating the hierarchical relationships between groups. The cluster visualization in Figure 6 shows that these groupings align closely with those observed in K-means clustering.

The cluster summaries in Table 3 further highlight differences between groups. For example, one cluster is characterized by high acousticness and low energy, while another shows higher speechiness and moderate energy. These patterns are consistent with those seen in Table 2, reinforcing the stability of the clustering structure across methods. Although hierarchical clustering is limited by computational constraints, its outputs provide strong validation for the cluster groupings.

hier_cluster	danceability	energy	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo	avg_popularity
0	0.632698	0.650666	-6.868407	0.080274	0.228058	0.019588	0.161965	0.481831	123.049716	32.349815
1	0.372131	0.194106	-19.940042	0.050563	0.865837	0.602182	0.135749	0.246613	111.457622	20.747012
2	0.662617	0.614557	-9.01784	0.367374	0.281089	0.033042	0.190061	0.526597	119.030035	27.61413
3	0.559373	0.68471	-9.791126	0.06976	0.146701	0.773422	0.198807	0.375805	119.168701	23.479885
4	0.499029	0.635824	-9.155884	0.138466	0.315469	0.096262	0.691363	0.476326	122.91322	25.908537

Table 3: Hierarchical Clustering Audio Features Breakdown

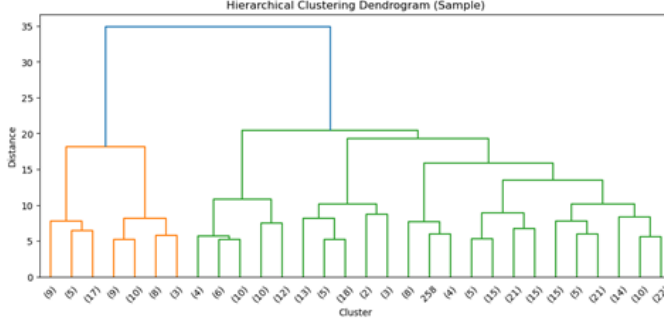


Figure 5: Dendrogram illustrating hierarchical cluster formation.

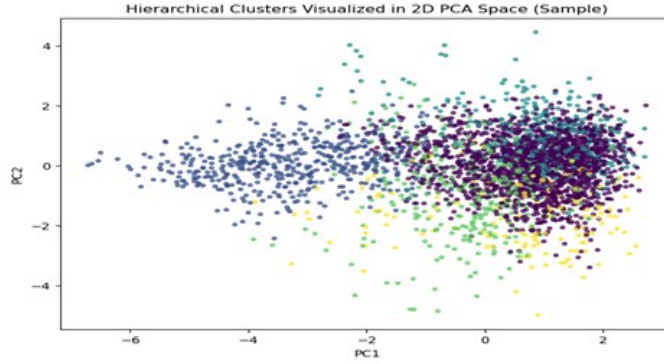


Figure 6: Hierarchical clustering results on sampled PCA data

VI. PyTorch

The autoencoder model introduces a nonlinear approach to feature representation, and its training progress is shown in Figure 8, where the decreasing loss indicates successful learning. The trained autoencoder produced a latent feature representation that reduced the dimension to only 3 main components, meaning each song was compressed into a 3-dimensional representation capturing its most important underlying characteristics. The latent feature representation, summarized in the output and visualized in Figure 9, shows improved cluster separation compared to PCA-based methods.

The cluster summaries in Table 4 reveal distinct groupings in the latent space. For example, one cluster exhibits high energy and tempo, while another shows high acousticness and lower energy, similar to patterns observed in earlier methods. However, the separation between clusters is more pronounced, indicating that the autoencoder captures deeper structure in the data. These results demonstrate that nonlinear transformations can enhance clustering performance by uncovering relationships not visible in the original or PCA-transformed space.

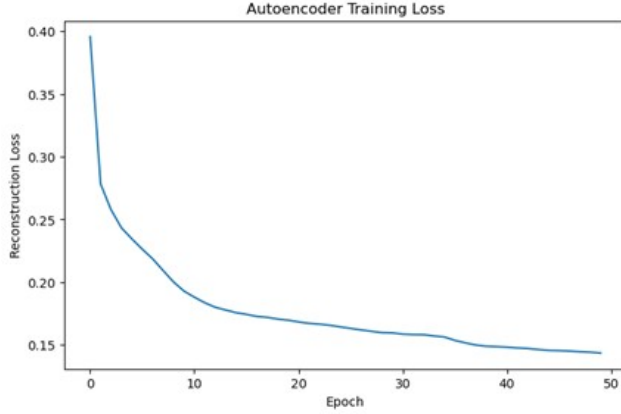


Figure 8: Autoencoder Training Loss

ae_cluster	danceability	energy	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo	avg_popularity
0	0.681784	0.603374	-9.145494	0.358956	0.275515	0.021814	0.25121	0.521356	108.873568	28.163199
1	0.652274	0.666114	-6.641775	0.100952	0.17426	0.017399	0.134826	0.446482	125.264062	32.388211
2	0.549742	0.710969	-7.801059	0.06613	0.13903	0.495118	0.155163	0.434163	142.882379	26.646882
3	0.441404	0.23302	-17.46832	0.053506	0.818728	0.447166	0.125664	0.315639	104.69516	23.338835
4	0.559748	0.695245	-8.093952	0.073149	0.27007	0.129663	0.446578	0.528714	113.167928	28.117828

Table 4: Autoencoder Audio Features Breakdown

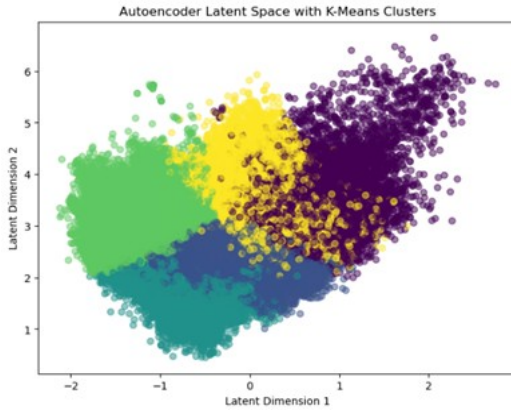


Figure 9: Clusters in autoencoder latent space

VII. Results

The clustering methods applied in this project revealed meaningful structure within the Spotify dataset and showed that songs can be grouped using measurable audio characteristics rather than traditional genre labels alone. Throughout all of the approaches, clusters were formed based on combinations of danceability, energy, acousticness, instrumentalness, tempo, and valence, indicating that multiple dimensions of song structure influence grouping behavior.

The baseline K-Means model, applied to PCA-transformed data with five clusters, produced interpretable segments of songs. For example, one cluster contained songs with relatively high danceability, energy, and valence, while another consisted of low energy songs with high acousticness and instrumentality. These findings suggest that K-Means successfully separated songs into practical listening profiles such as energetic dance tracks, acoustic songs, and instrumental focused music.

Hierarchical clustering was then used as a secondary method to validate whether similar groupings emerged under a different clustering framework. Because of computational limitations, this method was applied to a representative sample of the dataset. Even with sampling, the resulting cluster summaries aligned closely with the K-Means results, providing evidence that the overall cluster structure was stable across methods.

The strongest performance was achieved through the PyTorch autoencoder combined with K-Means clustering on the learned latent space. By compressing the original features into a three dimensional nonlinear representation, the autoencoder captured deeper relationships among songs and produced clearer cluster separation. This was supported by internal validation metrics. The autoencoder approach achieved the highest Silhouette Score (0.2904) and lowest Davies-Bouldin Index (1.1034), outperforming both PCA , K-Means, and hierarchical clustering.

Overall, the results indicate that nonlinear feature learning can improve clustering quality, while simpler classical methods still provide interpretable and computationally efficient baselines. In addition, several clusters showed differences in average popularity, suggesting that certain combinations of musical traits may be more strongly associated with listener engagement.

VIII. Challenges

There were some challenges that we encountered throughout this project. Our original plan was to collect data directly from the Spotify API. However, access limitations and data retrieval issues made that approach impractical within the project timeline. As a result, we transitioned to the “Spotify Audio Features April 2019” dataset, which provided a reliable and sufficiently large source of song observations for analysis.

The second challenge we encountered was the computational scale of the dataset. The dataset has more than 130,000 songs before filtering and more than 111,000 songs after preprocessing. While beneficial for analysis, this size increased runtime for clustering methods, especially hierarchical clustering. To address this, hierarchical clustering was performed on a representative sample of 3,000 songs.

Another challenge involved evaluating unsupervised learning models, since clustering does not provide known labels or a single correct solution. Therefore, internal metrics such as Silhouette Score and Davies-Bouldin Index were used to compare methods objectively. Selecting the

number of clusters also required balancing statistical guidance from the elbow method with practical interpretability.

This project also has some limitations. Spotify audio features measure technical song characteristics, but they do not capture lyrics, cultural context, listener demographics, or marketing effects. Thus, the clusters represent sound based groupings rather than complete measures of popularity or artistic value.

From a societal perspective, this type of analysis can improve recommendation systems and music discovery. However, it may also reinforce popularity bias by favoring already successful artists, highlighting the need for fairness and diversity in data-driven recommendation systems.

IX. Conclusion

This project demonstrated that machine learning techniques can successfully uncover hidden patterns within large scale music data. Using Spotify audio features, we applied PCA, K-Means clustering, hierarchical clustering, and a PyTorch autoencoder to identify natural groupings of songs based on measurable characteristics.

The results showed that songs consistently cluster according to combinations of energy, danceability, acousticness, tempo, and related features. While classical methods such as K-Means and hierarchical clustering produced meaningful and interpretable groups, the autoencoder model achieved the strongest performance by learning a nonlinear latent representation of the data.

These findings support the idea that music can be analyzed beyond traditional genre labels through quantitative feature profiles. These types of approaches have practical value in recommendation systems, playlist generation, and market segmentation within the music industry.

Future work could improve the analysis by incorporating lyrics, release year, or listener behavior data. Additional deep learning architectures may also produce stronger clustering performance. Overall, this project shows how modern analytics can transform entertainment data into actionable insights while demonstrating the value of combining statistical learning with deep learning methods.

X. Division of Labor

Research was a collaborative process centered around precise research objectives and identifying a dataset to meet requirements. The team developed a comprehensive project timeline to define key milestones and ensure rigorous methods were used in addressing core research questions. We ensured this timeline included the required deliverables, a mid-semester progress

presentation and the final report. While the technical execution was a collaborative effort, the responsibilities for the deliverables were divided.

- Foundations and Data Preprocessing: Hadleigh Dumale focused on introducing the project and dataset while discussing the technical steps involved in data cleaning and preparation.
- Statistical Data and Modeling: Spencer Andries was responsible for presenting the exploratory data analysis and discussion on the statistical models based on the datasets characteristics.
- Results and Critical Synthesis: Joshua De Leon headed the reporting of analytical results, discussing any technical constraints, and formulating a direction for any future research.